THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. How many non-square numbers lie between 9^2 and 10^2? 1

(a) 17 **(b)** 19 **(c)** 18 **(d)** 9

2. The value of $\frac{1}{2^{-3}}$ is: 1

(a) $\frac{1}{8}$ **(b)** 8 **(c)** 6 **(d)** -8

3. Which of the following is NOT a correct Roman numeral? 1

(a) IV **(b)** XL **(c)** IX **(d)** IC

4. The angles of a quadrilateral are in the ratio 1:2:3:4. The largest angle is: 1

(a) 132° **(b)** 156° **(c)** 108° **(d)** 144°

5. Is 3465 divisible by 11? 1

(a) Only if the number is a perfect square **(b)** Yes

(c) No **(d)** Only if the last digit is even

6. The value of $15^2 - 14^2$ is: 1

(a) 28 **(b)** 29 **(c)** 15 **(d)** 30

7. The value of $(-3)^4$ is: 1

(a) -81 **(b)** 12 **(c)** 85 **(d)** 81

8. The decimal value of 243_5 is: 1

(a) 73 **(b)** 71 **(c)** 78 **(d)** 75

9. In a parallelogram ABCD, $\angle A = (2x + 40)^\circ$ and $\angle C = (3x + 10)^\circ$. The value of x is: 1

(a) 30 **(b)** 31 **(c)** 29 **(d)** 35

10. Is 4518 divisible by 6? 1

(a) Yes **(b)** Only if the number is a perfect square

(c) No **(d)** Only if the last digit is even

11. The triangular number T_8 (sum of 1 to 8) is: 1

(a) 64 **(b)** 44 **(c)** 36 **(d)** 28

12. The standard form of 60000000 is: 1

(a) 60×10^7 **(b)** 6×10^6 **(c)** 6×10^7 **(d)** 6×10^8

13. The number 87 written in base 5 is: 1

(a) 223 **(b)** 321 **(c)** 322 **(d)** 323

14. In rhombus ABCD, $\angle ABC = 70^\circ$. The angle $\angle OBC$ (O is the point where the diagonals meet) is: 1

(a) 110° **(b)** 70° **(c)** 55° **(d)** 35°

15. For the number 2538, (largest 4-digit arrangement of its digits) - (smallest arrangement) is: 1

(a) 6183 **(b)** 6264 **(c)** 6084 **(d)** 6174

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Express 100 as the sum of consecutive odd numbers starting from 1. 2

17. Express $(2^3)^4 \times 2^{-5}$ as a single power of 2. 2

18. Convert 314_5 to a decimal number. 2

19. The angles of a quadrilateral are in the ratio 1:3:5:3. Find the smallest angle. 2

20. Use the divisibility rule for 11 to test 6578. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Show that $T_5 + T_6$ is a perfect square. 3

22. Express each as a single power of 2: 3

(i) $2^5 \times 2^{-2}$

(ii) $(2^3)^{-2}$

(iii) $2^4 \div 2^{-3}$.

23. Convert 137 to base 5 and check your answer by converting back to decimal. 3

24. The angles of a quadrilateral are in the ratio 1:2:3:4. Find all the angles. 3

OR

Check whether 35640 is divisible by 8 and by 11. 3

25. Check whether $(22, 120, 122)$ is a Pythagorean triplet. 3

OR

Find the value of $\left(\frac{2}{3}\right)^{-1} \times \left(\frac{2}{3}\right)^{-1}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Verify that 1729 can be written as the sum of two cubes in two different ways: $1^3 + 12^3$ and $9^3 + 10^3$. 5

OR

Find x if 5

(i) $2^x \times 2^3 = 2^{11}$

(ii) $(2^x)^4 = 2^{20}$

(iii) $2^{x-2} = \frac{1}{2^3}$.

27. Add the binary numbers 1101_2 and 111_2. Verify your answer by converting to decimal. 5

OR

(i) Show that the sum of the angles of a quadrilateral is 360°. 5

(ii) Three angles of a quadrilateral are 78°, 92° and 104°. Find the fourth angle.

END OF QUESTION PAPER **Total: 50 Marks**

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